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Case Title: Manson

Case Overview

Title: Manson Case

Background: I was retained by the court to conduct a forensic examination of Jake Manson's hard disk, a 26-year-old male currently on parole. Jake has prior convictions involving theft and misdemeanor drug charges. More recently, he was arrested for trafficking stolen and illegally purchased firearms. As part of a plea agreement, Jake avoided jail time but was placed under strict parole conditions.

Under the terms of his parole, Jake is prohibited from using a computer to engage in any activity related to the buying or selling of firearms. He was also cautioned that visiting firearm-related websites could raise suspicion and potentially constitute a violation.

During a routine visit, Jake's parole officer used HELIX to conduct a preliminary review of his hard drive. Upon discovering images of firearms, she reported a suspected parole violation. Jake disputes the allegation. He admits to occasionally browsing firearm-related websites but claims he is no longer involved in selling guns. He further states that any files related to firearm sales would have existed before his earlier arrest. Jake also notes that the computer is shared with his girlfriend and is accessible to visiting friends.

The court has requested a detailed forensic analysis of the hard disk to determine whether the firearm-related material is consistent with casual browsing or suggests deliberate and ongoing sales activity. Accurate timelines and user attribution are critical to this determination.

Objective: The objective of this report is to analyze Jake Manson's hard disk to determine whether he violated the terms of his parole by engaging in activity related to the buying or selling of firearms.

Scope: This investigation is limited to the forensic analysis of the provided hard disk image using FTK. Antivirus software was also used to perform a scan of the system for malicious files. No other external forensic tools were utilized. No external devices, network captures, or additional systems were examined.

The analysis focuses strictly on digital artifacts recovered from the hard disk, including browser history, Index.dat records, email files, temporary internet files, and locally stored documents relevant to firearm-related activity.

Methodology: To investigate the suspected parole violation, I followed standard digital forensic procedures beginning with documentation and preservation. A chain of custody form was completed before analysis. A verified working copy of the forensic image

Manson-HDD.E01(MD5:75352a41ac72f670b7673cf9634b7eee) was created, and the hash value of the copy was compared to the original acquisition hash to confirm integrity. The image was then scanned using antivirus software to ensure no malware was present that could artificially generate or alter firearm-related artifacts.

The analysis was conducted using **AccessData FTK version 8.2**. I first identified the file system type to understand how data would be structured and recovered. The system was determined to use FAT32, and further examination confirmed the operating system as Windows 98. Identifying the operating system was critical because Windows 98 handles user profiles and system artifacts differently than modern Windows systems. In particular, Windows 98 lacks the detailed user registry structures and account logging found in later versions, which impacts attribution analysis.

After identifying the operating environment, I focused on parsing Index.dat files, as these contain internet browsing history, cached records, and web activity relevant to firearm-related research, buying, selling, or email communication. The Index.dat records confirmed the system was operating as Windows 98, as many entries were associated with the default user profile "98user." Frequent web activity was observed, including firearm-related websites visited multiple times per day, consistent with admitted browsing behavior.

The browsing artifacts also revealed patterns consistent with multiple users accessing the machine. Searches related to jewelry, vacations, and leaving the United States were primarily observed during evening hours, while firearm-related browsing activity frequently occurred around 11:00 AM on multiple days. This time-based separation was documented to assist in user attribution analysis.

Following initial artifact review, I conducted targeted index searches within FTK using firearm-related keywords including: “p90,” “glock,” “glock 18,” “ammo,” “customer,” “gun,” “email,” “hotmail,” and “msn.” I also searched for specific file extensions, including .csv, .xls, .jpeg, .gif, and .png, to identify downloaded documents and images. These searches revealed multiple firearm images that had been deliberately downloaded and stored in:

- /My Documents/Merchandise/
- /Windows/Profiles/Jake/My Documents/Merchandise/

These directory structures indicate intentional organization rather than temporary internet caching.

Further examination revealed that a user profile named “Jake” existed on the system and had downloaded firearm images into that account’s Documents folder. Another profile named “Marilyn” was identified. The browsing history associated with Marilyn closely mirrored the activity of the default “98user,” particularly during evening hours. This supported the claim that multiple individuals had access to the system.

Additional artifacts recovered included a spreadsheet titled “**sales-05.xls**”, which contained customer names, product names, quantities, and prices. The product entries referenced firearm manufacturers and models. A separate document titled “**Things to do.doc**” contained a list of firearm-related materials, including gun safes, ammunition, and oil kits. These locally created documents were examined for timestamps and correlated with browsing activity.

A total of six firearm images were identified as downloaded to user document folders under both “98user” and “Jake.” These downloads were distinguished from temporary Internet files and were treated as deliberate file acquisition.

Email artifacts were also examined. Two relevant email files were recovered associated with the account “**guns1965@hotmail.com**”. One message referenced communication from “**eddie_zeddie@yahoo.com**”, and the email header data associated the guns1965 account with Jake Manson. Additionally, the message content included the statement, “Jake, I’m ready. Let’s make some deals...,” which provides contextual support for attribution analysis.

Due to the limitations of Windows 98, definitive user attribution is more complex than on modern operating systems because detailed per-user registry hives and login artifacts are limited. Therefore, attribution in this case relied on correlation between user profile directories, file creation timestamps, email headers, and browsing patterns.

All recovered evidence was documented with corresponding evidence numbers and hash values to maintain integrity and evidentiary reliability throughout the investigation.

Executive Summary

Case Title: *Manson Case*

Background: Jake Manson, a 26-year-old male currently on parole, is prohibited from using a computer to engage in firearm-related buying or selling. During a routine compliance review, his parole officer identified firearm images on his hard drive and reported a suspected violation.

Jake disputes the allegation, stating that he only casually browses firearm-related websites and that the computer is shared with others. The court requested a forensic examination to determine whether the recovered artifacts reflect casual browsing or deliberate sales activity, with particular focus on timeline and user attribution.

Objective: The objective of this investigation is to analyze Jake Manson's hard disk to determine whether he violated the terms of his parole by engaging in firearm-related buying or selling activity. This report evaluates digital artifacts, establishes relevant timelines, and assesses user attribution to determine whether the activity reflects casual browsing or deliberate sales behavior.

Questions to be answered:

1. **What operating system and file system were in use, and how do they affect artifact interpretation and user attribution?**
2. **Did Jake Manson engage in activity related to the buying or selling of firearms?**
3. **Do the recovered artifacts reflect casual firearm-related browsing or deliberate, organized sales activity?**
4. **When did the firearm-related activity occur, and what patterns exist within the timeline?**
5. **Who was likely using the computer at the time the firearm-related activity occurred?**
6. **What limitations exist in the available evidence, and how do those limitations affect definitive user attribution?**

Scope: This investigation is limited to the forensic examination of the hard disk image provided in this case. All findings are derived solely from artifacts recovered from that image using AccessData FTK 8.2. The image was also scanned using Avast Antivirus to ensure that no malicious software was present that could influence user activity or artifact generation.

No external devices, network logs, cloud accounts, or additional systems were analyzed. The review was confined to locally stored digital artifacts, including Index.dat records, browsing history, email files, downloaded images, spreadsheets, and user-created documents relevant to firearm-related activity.

Evidence: The evidence analyzed in this case consists of the forensic hard disk image **Manson-HDD.E01**(MD5:75352a41ac72f670b7673cf9634b7eee). All findings presented in this report originate from artifacts recovered from this image and were preserved with associated evidence identifiers and hash values to maintain integrity.

Findings: The analysis revealed multiple important findings that help form a complete picture of the incident:

- glock18.jpeg — Item #2059 — MD5: 5812486f571fea3d0a17b8cbe99533a6
- aug_r.jpeg — Items #2060 and #6196 — MD5: 1119116bb8a53b3e79e2ff213a9d8410b
- FN-5-7.gif — Item #2061 — MD5: 2ac30e7c02e7f513220f981096bd65e5
- p90.jpeg — Items #2062 and #6198 — MD5: 1618f0f49417b60d0ba590400f187597
- sales-05.xls — Item #5960 — MD5: d087b465b3ec5588000b3b977e4c78bc
- Things to do.doc — Item #5961 — MD5: c96abefc43ba8606d7637a0c1c8f381d
- getmsg[1].html — Item #4722 — MD5: 35563d1b6a92af921c3eb17b576f469
- compose[1].html — Item #5415 — MD5: a525a5c769578c9552141567fa8e796
- DC0.doc — Item #1353 Deleted document recovered — identical MD5 hash to "Things to do.doc"
- Jewelry image —Item #2253- recovered from /root/My Documents/Jewelry/ - MD5:f56c81f31cc143600c4813ee294759b5

Investigative Narrative: As the forensic analyst assigned to examine the digital evidence in the Manson case, I conducted a structured forensic review of the hard disk image provided to the court. The primary tool used during this investigation was AccessData Forensic Toolkit (FTK) version 8.2, which was used to analyze the forensic image file **Manson-HDD.E01**(MD5:75352a41ac72f670b7673cf9634b7eee).

The examination focused on identifying artifacts related to internet usage, downloaded files, locally created documents, and email communications stored on the system. Particular attention was given to Index.dat records, browsing history artifacts, downloaded image files, spreadsheets, documents, and email artifacts located within user profile directories. Temporary internet files were reviewed during the analysis process but were not emphasized in the findings, as cached browsing data alone does not demonstrate deliberate user intent.

The artifacts examined were evaluated based on file paths, timestamps, user directories, and associated metadata in order to reconstruct relevant system activity. This process allowed for the identification of files that were intentionally downloaded or stored on the system rather than artifacts generated automatically through normal web browsing.

The purpose of this investigation was to document the digital evidence present on the hard disk and evaluate whether the recovered artifacts reflected casual browsing behavior or more deliberate storage and organization of firearm-related material. All observations in this report are based solely on artifacts recovered from the forensic image and are presented without assumption or speculation.

Phase 1: System Identification:

The first step in the examination was determining what operating system and file system the computer was using. This was done by examining system artifacts and directory structures within the forensic image using FTK. Through this process, it was determined that the system was using the FAT32 file system and was running Windows 98. Additional confirmation came from browsing artifacts located within Index.dat files. Many of the Index.dat entries referenced the default user “98user,” which is commonly associated with Windows 98 systems.

Purpose and What it Reveals:

Determining the operating system and file system is an important early step in a forensic investigation because it helps explain how the system stores user activity and other artifacts. Different operating systems organize files, browsing history, and user information differently, so identifying the environment helps guide where relevant evidence can be located during the investigation. In this case, identifying the system as Windows 98 is important because these systems do not maintain detailed user login records like modern operating systems. Windows 98 commonly uses default profiles such as “98user,” and user attribution is limited compared to newer systems. Because of this, activity cannot always be directly tied to a specific user account, and attribution must rely more heavily on other artifacts such as file paths, browsing activity, user directories, and correlations between recovered evidence.

Phase 2: Web Activity Analysis

After determining that the system was running Windows 98, it became clear that direct user attribution would be more difficult than on modern Windows operating systems. Because of this limitation, the investigation relied more heavily on correlating and corroborating artifacts and timelines in order to better understand how the system was being used. As part of this process, the Index.dat files were analyzed to examine browsing activity on the machine.

All Index.dat files located on the system were manually parsed during the examination. Each artifact identified during this process was assigned an evidence number through FTK and associated with a hash value to preserve integrity. These artifacts are documented and included in the timeline phase of this report. The browsing artifacts recovered from the Index.dat files revealed a large amount of internet activity on the system.

The browsing history showed several firearm-related searches occurring primarily during the morning hours. These searches included specific firearm models such as “Glock 18,” “FN 57,” and “Bullpup.” The browsing history also showed frequent visits to firearm-related websites, including fnherstal.com and world.guns.ru. Other firearm-related browsing activity was also observed, although some of these visits were more general in nature and not directly related to the buying or selling of weapons. These artifacts were documented through FTK bookmarks for reference and further review if necessary.

The information recovered from the Index.dat files also helped guide additional investigative steps. The firearm models, websites, and keywords identified in the browsing history provided useful terms for index searches of the forensic image. These searches helped locate additional artifacts on the system, including downloaded firearm images and other related files stored within user directories.

The Index.dat artifacts also revealed frequent email login and logout activity on the system. In contrast to the firearm-related activity that occurred primarily in the morning hours, browsing activity occurring later in the evening showed searches related to jewelry, vacations, and travel destinations outside of the United States.

During the examination of user directories, an additional image artifact related to jewelry was identified within the Windows 98 default user directory located at:

`/root/My Documents/Jewelry/`

This artifact appears unrelated to firearm activity and may be associated with the non-firearm browsing behavior observed during evening hours in the Index.dat records. The presence of unrelated artifacts within different user directories further supports the possibility that multiple individuals had access to the system.

Finally, the Index.dat artifacts also revealed evidence of additional user profiles on the machine. In addition to the default “98user” profile commonly associated with Windows 98 systems, the browsing records referenced two other user accounts named “Jake” and “Marilyn,” indicating that multiple users may have had access to the computer.

Purpose and What it Reveals:

Analyzing the Index.dat files was important because these artifacts provide a record of browsing activity that occurred on the system. This information helps reconstruct how the computer was being used and what types of content were being accessed. In this case, the browsing history revealed repeated searches for specific firearm models and visits to firearm-related websites, which helps establish patterns of firearm-related activity on the system. The browsing artifacts also provided investigative leads that guided further searches of the forensic image, allowing additional relevant artifacts to be identified. At the same time, the presence of other browsing

topics and multiple user profiles suggests that the computer may have been used by more than one individual, which is an important factor when evaluating attribution in this investigation.

Phase 3: Downloaded File Analysis

After identifying firearm-related searches and websites through the Index.dat analysis, index searches were conducted across the forensic image using keywords that had been observed in the browsing artifacts. These searches included firearm models and related terms such as Glock, Glock 18, P90, fn, and other firearm-related keywords. The purpose of these searches was to determine whether firearm-related files had been intentionally downloaded or stored on the system.

Through this process, several firearm image files were located within user directories on the system. The images were not located within temporary internet folders but were instead stored within user document directories. Specifically, these images were located in directories such as:

/My Documents/Merchandise/<image_name>

/Windows/Profiles/Jake/My Documents/Merchandise/<image_name>

The presence of the “Merchandise” folder within these directories indicates that these files were intentionally saved and organized rather than automatically generated through normal web browsing activity.

The firearm image artifacts identified during the examination include the following files:

glock18.jpeg — Item #2059 (*Item III in appendix*)

aug_r.jpeg — Item #2060 (*Item I in appendix*)

FN-5-7.gif — Item #2061 (*Item II in appendix*)

p90.jpeg — Item #2062 (*Item III in appendix*)

aug_r.jpeg — Item #6196 (duplicate of Item #2060, identical MD5 hash) (*Item I in appendix*)

p90.jpeg — Item #6198 (duplicate of Item #2062, identical MD5 hash) (*Item IV and V in appendix*)

17668595_FL_LRG.jpeg — Item #2253 (*Item IX in appendix*)

DC0.doc — Item #1353 (*Item X in appendix*)

These six artifacts represent four unique firearm images, with two of the files appearing as duplicates that share identical MD5 hash values. The images correspond to specific firearm models, including the Glock 18, Steyr AUG, FN Five-sevenN, and FN P90.

Further examination of the file paths revealed that some of these images were located within the user profile directory associated with the account name “Jake.” This is significant because the user account “Jake” had previously been identified through browsing artifacts recovered from the Index.dat files during earlier phases of the investigation.

Purpose and What it Reveals:

Examining downloaded artifacts is important because files stored within user document directories typically indicate deliberate user action rather than normal browsing activity. The system automatically creates temporary Internet files during browsing, and these do not necessarily indicate that a user intentionally saved the content. In contrast, files located within user document folders generally require a user to download and store them intentionally. In this case, the presence of multiple firearm images stored within a folder labeled “Merchandise” suggests a level of organization and intentional storage of firearm-related material. The correlation between the firearm-related browsing activity identified in the Index.dat files and the downloaded images strengthens the timeline of firearm-related activity on the system. It indicates that the browsing activity may have led to the deliberate storage of these files.

Phase 4: Document Analysis

After identifying the downloaded firearm images stored within the user directories, additional index searches and file reviews were conducted to determine whether any locally stored documents related to firearm activity were present on the system. These searches focused on common document file types such as .xls, .csv, and .doc to identify files that may contain written notes, records, or other organized information.

Through this process, two documents were identified within the user profile associated with the account name **Jake**. Both files were located within the directory:

/Windows/Profiles/Jake/My Documents/

The two documents identified were “**sales-05.xls**” and “**Things to do.doc**”. Because these files were located within the Jake user document directory and were not associated with temporary internet files or browser cache artifacts, this indicates that they were intentionally stored on the system.

During the document analysis phase, an additional artifact, identified as **DC0.doc**, was recovered from the system. This file represents a deleted version of the document titled *Things to do.doc*. Examination of the file metadata revealed that **DC0.doc and Things to do.doc share identical MD5 hash values**, confirming that they are the same file. This indicates that the document had previously been deleted and later recovered through forensic analysis. The recovery of deleted artifacts demonstrates the ability of forensic tools to recover evidence that may have been intentionally removed from the system.

The file “**sales-05.xls**”(Item VI in the appendix) is a spreadsheet that contains structured information, including product names, customer names, quantities, prices, and totals. The entries within the spreadsheet reference firearm manufacturers and models. The format of the spreadsheet suggests that firearm-related items were being tracked in an organized manner.

The second document, “**Things to do.doc**”(Item VII in the appendix), contains a written list of items, including purchasing a gun safe from Albert, acquiring gun cleaning kits, and purchasing oil and ammunition. The document appears to be written as a task or planning list and was stored within the Jake user document directory along with the other recovered artifacts.

Both documents were assigned evidence numbers through FTK and associated with MD5 hash values to preserve the integrity of the evidence.

Purpose and What it Reveals:

Examining locally stored documents is important because files saved within user directories generally indicate deliberate user activity rather than artifacts automatically generated by the system. Unlike browsing history or cached web files, documents such as spreadsheets and text files typically require a user to intentionally create or save them. In this case, the presence of a spreadsheet containing firearm-related product information and a document listing firearm-related materials provides additional context for how the system may have been used. Because both files were located within the Jake user profile directory, they also contribute to the attribution analysis later in the report.

Phase 5: Email Analysis

During the examination of the system artifacts, email-related files were also identified. These artifacts were located during the review of browsing and file artifacts and were examined to determine whether any communications related to firearm activity were present on the system.

Two email artifacts were identified during the investigation. These files were recovered as HTML artifacts representing saved email content and composed messages. The files were identified as “**getmsg[1].html**” and “**compose[1].html**”, and both were assigned evidence

numbers through FTK. The file “getmsg[1].html” was assigned **Item #4722** with an MD5 hash value of **35563dd1b6a92af921c3eb17b576f469**. The file “compose[1].html” was assigned **Item #5415** with an MD5 hash value of **a525a5c7c69578c9552141567fa8e796**.

Both artifacts were recovered from the following directory: *(Item VIII in the appendix)*

/WINDOWS/Temporary Internet Files/Content.IE5/

The recovered email artifacts revealed the presence of the email account **guns1965@hotmail.com** on the system. One of the recovered messages contained communication from the email address **eddie_zeddie@yahoo.com** directed toward this account. Within the message content, the sender refers to an individual named “Jake” and states, “*Jake, I'm ready. Let's make some deals...*”

These artifacts were recovered alongside other browsing and user artifacts previously identified on the system. The presence of email login and logout activity observed earlier within the Index.dat browsing artifacts also supports that email activity was occurring on the system.

Purpose and What it Reveals:

Examining email artifacts is important because they can provide evidence of communication occurring through the system and may help support attribution when other artifacts are limited. In this case, the recovered email artifacts reveal the presence of the email account **guns1965@hotmail.com** and show communication from another user referencing an individual named “Jake.” While this alone does not definitively prove ownership of the account, the reference to the name “Jake,” combined with the presence of a user profile with the same name on the system and other artifacts identified during the investigation, provides additional context that may support attribution when considered alongside the rest of the recovered evidence.

Phase 6: Timeline Correlation

Date	Time	Event	Evidence (Item / MD5)	Correlating Evidence Time	Correlation
9/26/2005	11:14:36 AM	Email composed from guns1965@hotmail.com	Item 4722 – 35563dd1b6a92af921c3eb17b576f469	11:14:42 AM	Email activity followed by firearm search
9/26/2005	11:14:42 AM	Google search "Glock 18"	Index.dat 5743	11:18:02 AM → 11:18:57 AM	Leads to Glock research and image artifact
9/26/2005	11:17:02 AM	Hotmail accessed	Index.dat 4032	11:14–11:20 session	Same browsing session
9/26/2005	11:18:02 AM	Remtek Glock model page accessed	Index.dat 5745	11:18:57 AM	Glock firearm image created
9/26/2005	11:18:57 AM	Glock 18 firearm image created	Item 2059 – 5812486f571fea3d0a17bcbe99533a6	11:14:42 AM → 11:18:02 AM	Matches Glock search and research
9/26/2005	11:19:42 AM	Google search "Glock 18"	Index.dat 5743	11:18:57 AM	Continued firearm research
9/26/2005	11:20:14 AM	Google search "bullpup aug-steyr"	Index.dat 4032	11:20:12 AM	Leads to AUG rifle artifact
9/26/2005	11:20:12 AM	AUG rifle image created	Item 2060 – 119116bb8a53b3e79e2ff213a9d8410b	11:20:14 AM	Matches bullpup search
9/26/2005	11:20:28 AM	gunsamerica.com accessed	Index.dat 4032	11:20 session	Firearm sales website
9/26/2005	11:21:13 AM	FN Herstal website accessed	TIF 5297	11:21:51 AM → 11:23:00 AM	Manufacturer of P90 firearm
9/26/2005	11:21:51 AM	FN 5-7 firearm image created	Item 2061 – 2ac30e7c2e7f513220f981096bd65e5	11:21:13 AM	Matches FN firearm browsing
9/26/2005	11:23:00 AM	P90 firearm image created	Item 2062 – 1618f049417b60d0ba590400f187597	11:21:13 AM	Correlates with FN Herstal site
9/26/2005	5:54:36 PM	Email referencing selling "merchandise"	Item 5415 – a525a5c7c69578c9552141567fa8e796	Same-day firearm artifacts	Merchandise likely refers to firearm images
10/31/2005	10:36:51 AM	sales-05.xls created	Item 5960 – d087b4653bc5588000b3b977e4c78bc	10:36:51 AM	Similar timestamp to planning document
10/31/2005	10:36:51 AM	things to do.doc created	Item 5961 – c96abefc43ba8606d7637a0c1c8f381d	10:36:51 AM	Similar timestamp to sales spreadsheet
10/31/2005	10:51:30 AM	User Jake login	Index.dat 6181	10:36 document creation	Same morning activity
10/31/2005	11:03:12 AM	Google search "Gun Pictures"	Index.dat 6181	Same browsing session	Continued firearm browsing

Timeline Analysis

The timeline above shows the sequence of firearm-related browsing activity, image artifacts, emails, and document creation that occurred on the system. On September 26, 2005, multiple searches were conducted for specific firearm models, including “Glock 18” and “bullpup aug-steyr.” Within minutes of these searches, firearm-related websites were accessed and image artifacts associated with those firearms were created. The close timing between the searches, website visits, and image creation suggests that these images were obtained during the same browsing session in which the firearms were being researched.

In addition to occurring within minutes of each other, the firearm images were also saved within the same directory on the system located at:

/Windows/Profiles/Jake/My Documents/Merchandise/

The fact that multiple firearm images were saved to the same directory indicates that the files were deliberately stored rather than simply remaining in temporary browser cache locations. The combination of closely aligned timestamps and a shared storage location suggests that these images were intentionally collected during the same browsing activity.

Additional activity during this time includes visits to firearm-related websites such as gunsamerica.com and the FN Herstal manufacturer website. Shortly after the FN Herstal site was accessed, image artifacts for FN firearms and a P90 firearm were created on the system. This pattern shows a sequence where firearm models were searched for, related websites were visited, and corresponding firearm images were created within a short time frame.

Later on the same day, an email artifact was identified that references selling “merchandise.” While the email does not explicitly mention firearms, the reference to merchandise is notable when considered alongside the firearm images stored in the Merchandise directory.

On October 31, 2005, two documents were created on the system titled *sales-05.xls* and *things to do.doc*. Both files were also located within the directory:

/Windows/Profiles/Jake/My Documents/

The spreadsheet contains fields for product names, customers, quantities, prices, and totals, while the second document contains a list of firearm-related supplies such as ammunition, gun cleaning kits, and a gun safe. Both documents share a similar timestamp, indicating they were likely created during the same session. Shortly after their creation, browsing artifacts show that the user account “Jake” was active on the system.

Overall, the timeline shows several clusters of firearm-related activity, including firearm searches, firearm website visits, firearm image artifacts saved to the same directory, email communication referencing merchandise, and the creation of documents related to firearm supplies and sales. These events occur within identifiable time periods and provide context for understanding how firearm-related materials were accessed and stored on the examined system.

Conclusion

This investigation examined the Manson hard disk image to determine the nature of firearm-related activity present on the system. Analysis of the file system, browsing artifacts, documents, email artifacts, and timeline revealed multiple searches for specific firearm models, visits to firearm-related websites, and the creation of firearm image artifacts. These images were created within minutes of the related searches and were stored together in the directory

/Windows/Profiles/Jake/My Documents/Merchandise/.

The firearm models researched during browsing activity, the firearm images downloaded to the system, and the products listed in the spreadsheet *sales-05.xls* all reference the same types of firearms, including the Glock 18, FN 5-7, P90, and bullpup-style rifles such as the Steyr AUG. Additional artifacts identified include a document titled *things to do.doc*, which lists firearm-related supplies such as ammunition, gun cleaning kits, and a gun safe.

Email artifacts were also recovered that reference selling “merchandise.” The email header and message content associate the email account **guns1965@hotmail.com** with the name “Jake,” and the message content references “Jake” directly.

The system was determined to be running Windows 98 on a FAT32 file system. Because of the limited user attribution capabilities of this operating system and the possibility of multiple individuals accessing the computer, it is not possible to definitively determine which individual performed each action identified during the investigation.

Appendix

Item I- aug_r.jpeg — 119116bb8a53b3e79e2ff213a9d8410b (Matching Hashes)

☐	aug_r.jpeg	2060	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchandise/aug_r.jpeg
☐	aug_r.jpeg	6196	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Profiles/Jake/My Documents/Merchandise/aug_r.jpeg
☐	aug_r[1].jpeg	5263	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/GHJGHMN/aug_r[1].jpeg

Loaded: 21 Filtered: 21 Total: 21 Highlighted: 1 Checked: 0 [Click to update size](#)

File Content

Hex Text Filtered Translation Natural



☐	aug_r.jpeg	6196	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Profiles/Jake/My Documents/Merchandise/aug_r.jpeg	JPEG	16.00 KB	12.78 KB	119116bb8a53b3e79e2ff213a9d8410b
☐	aug_r[1].jpeg	5263	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/GHJGHMN/aug_r[1].jpeg	JPEG	16.00 KB	12.78 KB	119116bb8a53b3e79e2ff213a9d8410b
☐	DC0.DOC	1353	doc	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/RECYCLED/DC0.DOC	Microsoft ...	20.00 KB	19.00 KB	c96abefc43ba8606d7637a0c1c8f381d

Loaded: 21 Filtered: 21 Total: 21 Highlighted: 1 Checked: 0 [Click to update size](#)

File Content

Hex Text Filtered Translation Natural



Item II - FSN1-5-7.gif — 2ac30e7c02e7f513220f981096bd65e5

☒	Name	It...	Ext	Path
☐	DC0.DOC	1353	doc	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/RECYCLED/DC0.DOC
☐	FSN1-5-7.gif	2061	gif	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchandise/FSN1-5-7.gif
☐	glock_all[1].jpeg	4619	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/CZSL2P...
☐	glock18.jpeg	2059	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchandise/glock18.jpeg
☐	index.dat	6558	dat	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Profiles/Marilyn/History/History.IE5/MSHist012
☐	nepal_pic[1].jpeg	5145	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/Z9STU4

Loaded: 21 Filtered: 21 Total: 21 Highlighted: 1 Checked: 0 [Click to update size](#)

File Content

Hex Text Filtered Translation **Natural**



Item III - glock18.jpeg — 5812486f571fea3d0a17b8cbe99533a6

☐	glock_all[1].jpeg	4619	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/CZSL2P...
☐	glock18.jpeg	2059	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchandise/glock18.jpeg
☐	index.dat	6558	dat	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Profiles/Marilyn/History/History.IE5/MSHist012
☐	nepal_pic[1].jpeg	5145	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/Z9STU4

Loaded: 21 Filtered: 21 Total: 21 Highlighted: 1 Checked: 0 [Click to update size](#)

File Content

Hex Text Filtered Translation **Natural**



1618f0f49417b60d0ba590400f187597(hashmatch)

[illegible]

Item VII- Things to do.doc — c96abefc43ba8606d7637a0c1c8f381d

Icon	Filename	Size	Type	Path	Encoding	Size	Size	Hash
doc	Things to do.doc	5961	doc	Manson-HDD.E01/Partition 1/NO NAME [FAT32]/[root]/WINDOWS/Profiles/Jake/My Documents/Things to do.doc	Microsoft ...	20.00 KB	19.00 KB	c96abefc43ba8606d7637a0c1c8f381d
jpeg	TR[1].jpeg	4676	jpeg	Manson-HDD.E01/Partition 1/NO NAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/2951U4N2/TR[1].jpeg	JPEG	8192 B	4182 B	544688f753db103e902f84c9cb2856b
jpeg	wheel280[1].jpeg	5056	jpeg	Manson-HDD.E01/Partition 1/NO NAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/CZSL2P41/wheel280[1].jpeg	JPEG	12.00 KB	11.76 KB	54121a6c2866dec127c7189aee58cf8f

Loaded: 21 | Filtered: 21 | Total: 21 | Highlighted: 1 | Checked: 0 | Click to update size

File Content

Hex | Text | Filtered | Translation | Natural

Things to do

Oil change.

Buy gun cleaning kits.

Pick up gun safe from Albert.

Buy ammo.

Item VIII- compose[1].html — a525a5c7c69578c9552141567fa8e796 getmsg[1].html — 35563dd1b6a92af921c3eb17b576f469

Icon	Filename	Size	Type	Path	Encoding	Size	Size	Hash
html	compose[1].html	5415	html	Manson-HDD.E01/Partition 1/NO NAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/GHJGHMN/compose[1].html	HTML	24.00 KB	23.69 KB	a525a5c7c69578c9552141567fa8e796

Loaded: 2 | Filtered: 2 | Total: 2 | Highlighted: 1 | Checked: 0 | Click to update size

File Content

Hex | Text | Filtered | Translation | Natural

Zed

Jake Manson <guns1965@hotmail.com> wrote:

Hey Zed,

Glad you got the e-mail account set up. I'll be sending some pics of merchandise soon. Fun stuff.

Jake

Don't just search. Find. Check out the new MSN Search!
<http://search.msn.click-url.com/go/onm00200636ave/direct/01/>

Yahoo! for Good
[Click here to donate](#) to the Hurricane Katrina relief effort.

File Content | Properties | Hex Interpreter

Icon	Filename	Size	Type	Path	Encoding	Size	Size	Hash
html	compose[1].html	5415	html	Manson-HDD.E01/Partition 1/NO NAME [FAT32]/[root]/WINDOWS/Temporary Internet Files/Content.IE5/GHJGHMN/compose[1].html	HTML	24.00 KB	23.69 KB	a525a5c7c69578c9552141567fa8e796

Loaded: 2 | Filtered: 2 | Total: 2 | Highlighted: 1 | Checked: 0 | Click to update size

File Content

Hex | Text | Filtered | Translation | Natural

From: Edgar Zed <eddie_zeddie@yahoo.com>

Sent: Monday, September 26, 2005 11:43 AM

To: Jake Manson <guns1965@hotmail.com>

Subject: Re: contact

Attachment: [Gallardo.jpg](#) (< 0.01 MB)

Jake,

I'm ready. Let's make some deals. I need some money soon.....

I want this car and it ain't cheap!

Zed

Jake Manson <guns1965@hotmail.com> wrote:

[Go to previous message](#) | [Go to next message](#) | [Delete](#) | [Inbox](#)

Item IX – DC0.doc (Deleted file – identical MD5 hash to Things to do.doc)

Things to do

Oil change.

Buy gun cleaning kits.

Pick up gun safe from Albert.

Buy ammo.

File Content									
Properties									
Hex Interpreter									
File List									
Display Time Zone: Eastern Standard Time (From local machine)									
☒	Name	Label	It...	Ext	Path	Category	P-Size	L-Size	MD5
☐	17668595_FL_LRG.jpeg		2253	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Jewelry/...	JPEG	12.00 KB	10.91 KB	f56c81f31cc143600c4813ee294759b5
☐	aug_r.jpeg		2060	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchan...	JPEG	16.00 KB	12.78 KB	119116bb8a53b3e79e2ff213a9d8410b
☐	FSN1-5-7.gif		2061	gif	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchan...	GIF	16.00 KB	13.40 KB	2ac30e7c02e7f513220f981096bd65e5
☐	glock18.jpeg		2059	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchan...	JPEG	12.00 KB	11.41 KB	5812486f571fea3d0a17b8cbe99533a6
☐	P90.jpeg		2062	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Merchan...	JPEG	8192 B	4533 B	1618f0f49417b60d0ba590400f187597
☐	DC0.DOC		1353	doc	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/RECYCLED/DC0.DOC	Microsoft ...	20.00 KB	19.00 KB	e96abefc43ba8606d7637a0c1c8f381d

Item X – Jewelry image recovered from /root/My Documents/Jewelry/

File Content

Hex Text Filtered Translation Natural



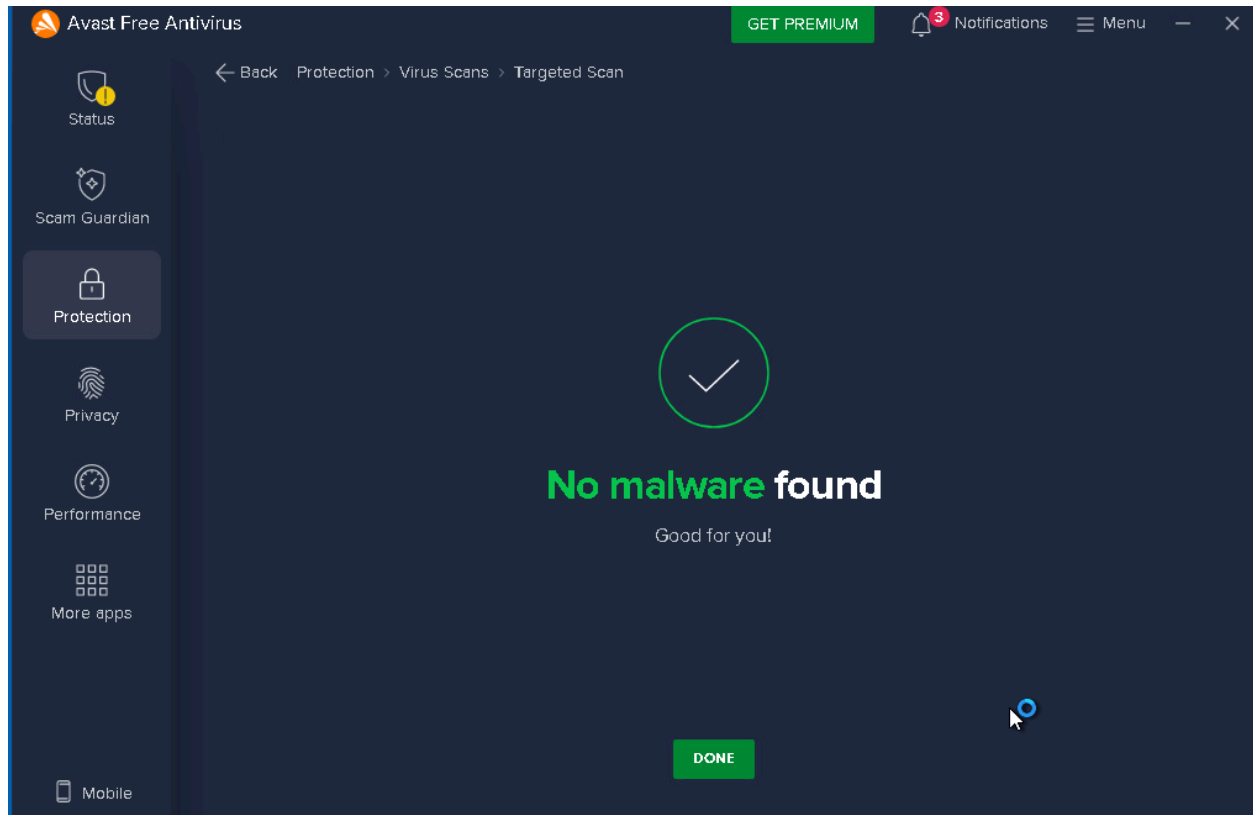
File Content Properties Hex Interpreter

File List

Display Time Zone: Eastern Standard Time ...

☒	Name	Label	It...	Ext	Path
☐	103902		1232		Manson-HDD.E01/Partition 1/NONAME [FAT32]/[unallocated space]/103902
☐	17668595_FL_LRG.jpeg		2253	jpeg	Manson-HDD.E01/Partition 1/NONAME [FAT32]/[root]/My Documents/Jewelry/17668595_FL_LRG.jpeg

Anti-Virus Scan (AVAST)



Manson Windows 98 machine hash-

